

Tendon injury: from biology to tendon repair

Geoffroy Nourissat, Francis Berenbaum and Delphine Duprez

Abstract | Tendon is a crucial component of the musculoskeletal system. Tendons connect muscle to bone and transmit forces to produce motion. Chronic and acute tendon injuries are very common and result in considerable pain and disability. The management of tendon injuries remains a challenge for clinicians. Effective treatments for tendon injuries are lacking because the understanding of tendon biology lags behind that of the other components of the musculoskeletal system. Animal and cellular models have been developed to study tendon-cell differentiation and tendon repair following injury. These studies have highlighted specific growth factors and transcription factors involved in tenogenesis during developmental and repair processes. Mechanical factors also seem to be essential for tendon development, homeostasis and repair. Mechanical signals are transduced via molecular signalling pathways that trigger adaptive responses in the tendon. Understanding the links between the mechanical and biological parameters involved in tendon development, homeostasis and repair is prerequisite for the identification of effective treatments for chronic and acute tendon injuries.

Nourissat, G. *et al.* *Nat. Rev. Rheumatol.* advance online publication 3 March 2015; doi:10.1038/nrrheum.2015.26

Introduction

Tendon is a unique form of connective tissue and is the component of the musculoskeletal system that links muscle to bone. This mechanosensitive tissue has specific mechanical properties that enable it to respond and adapt to loading transmitted by muscles. Tendon pathologies range from chronic injury to acute injury with partial or complete tendon rupture.^{1,2} Chronic tendon injury, or tendinopathy, is the most common overuse tendon injury. Tendinopathy is a condition characterized by pain and by impaired performance.³ The pathogenesis of tendinopathy is poorly understood and has been variously defined as a degenerative condition or as a failure of the healing process.^{4–6} Moreover, the role of inflammation in tendinopathy is not clearly established and is a matter of debate.⁷ The exact relationship between tendinopathy and tendon rupture remains unknown. However, it has been reported that tendinopathy could lead to tendon rupture.⁸ Partial or complete tendon ruptures interrupt tendon continuity and lead to diminution or loss, respectively, of transmitted forces, and potentially to loss of mobility. Following acute rupture, tendon undergoes a healing process, involving the successive steps of inflammation, extracellular matrix (ECM) formation and remodelling.^{1,2} The mechanisms underlying tendon healing are not fully characterized, but research involving animal and cellular models is ongoing.

Approximately 30% of general practice consultations for musculoskeletal pain are related to tendon disorders.³ The exact incidence of tendinopathy in the general population is difficult to assess as it is diagnosed as soft tissue pain, which is very common. Acute tendon injuries are also very common. The American Academy

of Orthopaedic Surgeons estimates that, in 2008, almost 2,000,000 people consulted a physician because of a rotator cuff problem.⁹ Tendon injury can affect people of all ages, and can impair the activity of young and old adults in their work environment or sports activities. In summary, tendon disorders are common, have a substantial effect on quality of life and represent an important economic burden on health-care systems.

Chronic or acute injuries can occur in any tendon, but often affect major tendons with high *in vivo* loading demands, such as the Achilles, patellar, rotator cuff and forearm extensor tendons.^{5,10,11} The response of tendon to abnormal mechanical loading has an important role in tendon injury. Chronic and acute tendon injuries are frequently related to physical requirements of employment and of sports.^{3,11} Mechanical forces are perceived by tendon cells as stimuli that are transduced via ECM, growth factors, receptors, intracellular pathways and transcription factors and converted to biochemical signals that elicit cellular responses.¹² Understanding the relationship between mechanical parameters, growth factors and transcription factors in tendon biology is crucial to identifying strategies and potential treatments for tendon repair. In this Review, we discuss the current understanding of the biological and mechanical parameters involved in tendon development, homeostasis and repair, and attempt to hierarchize these parameters in the context of tendon biology.

Tendon structure

As the anatomical structure that connects muscle and bone, tendon transmits muscle-contraction force to the skeleton to maintain posture or produce motion (Figure 1a). Ligaments have a similar structure to tendons, but link bone with bone, stabilize joints and

Service de chirurgie orthopédique et traumatologique (G.N.) and Service de rhumatologie (F.B.), INSERM UMR_S938, DHU i2B, Assistance Publique-Hopitaux de Paris, Hôpital Saint-Antoine, 184 rue du Faubourg Saint-Antoine, Paris 75012, France. Centre national de la recherche scientifique UMR 7622, IBPS Developmental Biology Laboratory, F-75005, Paris 5005, France (D.D.).

Correspondence to: F.B. francis.berenbaum@sat.aphp.fr

Competing interests

The authors declare no competing interests.

Key points

- Tendon is a mechanosensitive tissue
- Abnormal loading leads to tendon injuries
- Mechanical forces are converted to biochemical signals that elicit cellular responses by tendon cells
- Similar mechanical and biological signals are involved in tendon development, homeostasis and repair
- A better understanding of the interaction between forces, intracellular pathways and gene transcription in the context of tendon biology is needed
- Understanding mechanobiology in tendon development, homeostasis and repair is critical to designing therapies for tendon injury

usually operate at much lower ultimate tensile strength compared with tendons.¹⁰ Tendons have a hierarchical fibrillar arrangement, whereby triple-helical type I collagen molecules assemble into fibrils that, in turn, form fibres, fascicles and, ultimately, the tendon unit (Figure 1b). The collagen fibrils, considered to be the fundamental force-transmitting unit of the tendon, are densely arranged within the ECM, oriented parallel to the bone–muscle axis (Figure 1). A multitude of ECM molecules, including collagens, elastin, proteoglycans and glycoproteins, are involved in the tendon-specific fibrillogenesis of type I collagen.¹³

Type I collagen and associated ECM molecules are produced by tendon cells (tenocytes), which are fibroblast-like cells located inside collagen fibres and in the surrounding endotenon (Figure 1). Tendon stem cells (TSCs) have been isolated in cell cultures from human, rabbit, rat and mouse tendons and have regenerative properties different from those of bone-marrow-derived mesenchymal stem cells (MSCs).^{14,15} Mouse TSCs have been isolated from the tendon proper and from tendon sheaths but these cell populations differed from each other in cell-marker expression.¹⁶ The identification of TSCs has been based on colony-forming unit assays in cell cultures, and TSCs cannot currently be visualized in tendons *in situ*.

Type I collagen content and tendon cells are not homogeneous along the length of the tendon. Tendon is attached to muscle by the myotendinous junction and tendon is prolonged by septa and fascia into muscle. Tendon is attached to bone through a fibrocartilaginous tissue called the enthesis (Figure 1a).¹⁷ The enthesis is rich in type II collagen, which is produced by chondrocyte-like cells that are not present in other tendon regions.¹⁷ Consequently, the enthesis and midtendon have different cellular and molecular compositions, which lead to different responses to mechanical loading in normal and pathological conditions.

Type I collagen production in tendons

Because type I collagen is the most abundant component of the ECM not only in tendon, but in all soft tissues, studying tendon biology has not proved easy. The specific structure of tendon is determined not by the expression of type I collagen, but only by the specific parallel organization of type I collagen fibrils (Figure 1). To date, very little is known about the mechanisms driving the specific spatial organization of type I collagen fibres in tendon. However, data does exist on the biological factors

that regulate type I collagen production in tendon. Secreted growth factors such as transforming growth factor (TGF)- β and fibroblast growth factors (FGFs) are known to promote collagen expression and synthesis in tendon tissue during development and adult life.^{18–22}

In addition to growth factors, three transcription factors have been implicated in the expression of *Col1a1* and *Col1a2* (encoding collagen α -1[I] and collagen α -2[I], respectively) and other components involved in collagen fibrillogenesis: the basic helix-loop-helix transcription factor scleraxis, the homeobox protein Mohawk and the zinc-finger protein early growth response protein 1 (EGR1) (Table 1). In mice, genetic loss of scleraxis (*Scx*) leads to partial loss of *Col1a1* expression and complete loss of expression of *Col14a1* and *Tnmd* (encoding collagen α 1[XIV] and tenomodulin, respectively) in tendons and, consequently, tendon formation is impaired.²³ Mohawk and Egr1 have been shown to be important for *Col1a1* transcription in developing and adult mouse tendons.^{18,24–27} Mechanical forces are also involved in type I collagen protein synthesis in animal and human tendons: increased loading leads to increased collagen content in tendons, whereas decreased loading leads to decreased collagen content.^{21,28,29} How growth factors, transcription factors and mechanical forces are coordinated to control type I collagen production in tendons is, however, incompletely understood.

Chronic and acute tendon injuries

Tendon pathology involves disruption of the highly organized hierarchical collagen structure in tendons. Altered organization of type I collagen is observed in tendons of patients with genetic diseases affecting type I collagen fibrillogenesis, such as Marfan syndrome.³⁰ However, these genetic pathologies affect not only tendons, but all connective tissues containing type I collagen. Tendon-specific pathologies include mainly chronic and acute tendon injuries.

Chronic tendon injury, or tendinopathy, refers to the clinical symptoms including pain, focal tendon tenderness, decreased strength and movement in affected tendons.³ In contrast to partial or complete tendon rupture, tendinopathy occurs without macroscopic tearing of the tendon. Tendinopathy can be identified by the following histological characteristics: collagen fibril disorganization, increased proteoglycan and glycosaminoglycan content and increased noncollagenous ECM, hypercellularity and neovascularization.^{11,31–33} These cellular and molecular changes modify the mechanical properties of tendon and cause pain. Because the pathogenesis of tendinopathy is not fully understood, different hypotheses have been proposed, including degeneration and failed healing.^{3–6} Acute tendon injury refers to partial or complete rupture, which interrupts tendon continuity and can ultimately lead to loss of motion. Tendon rupture is followed by a natural healing process, although this healing is less efficient than in other components of the musculoskeletal system.^{1,2}

Chronic and acute tendon injuries are facilitated by many extrinsic and intrinsic factors.⁶ Common intrinsic

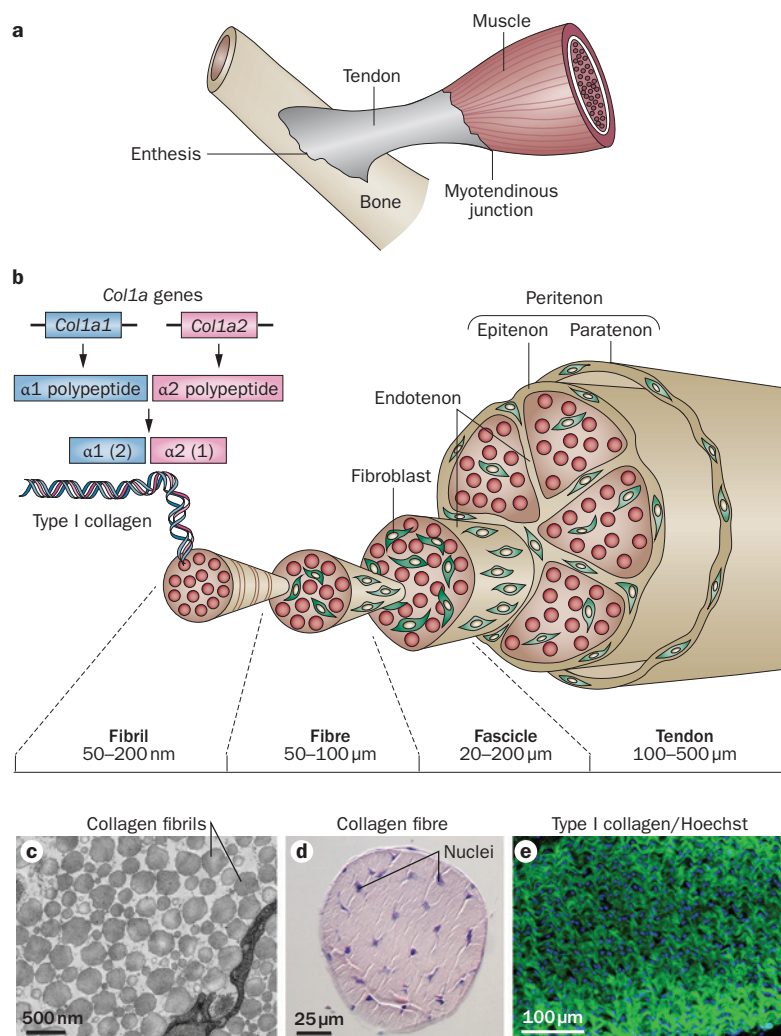


Figure 1 | Tendon architecture. **a** | Tendon links muscle to bone and is fixed to bone by the enthesis and to muscle by the myotendinous junction. Tendon is composed of spatially organized type I collagen fibres. **b** | Type I collagen, the major structural component of tendon, can be visualized at different scales. *Col1a1* and *Col1a2* code for collagen $\alpha 1(I)$ and $\alpha 2(I)$ polypeptides, respectively. Type I collagen triple-helical molecules containing two $\alpha 1(I)$ and one $\alpha 2(I)$ chains assemble into fibrils that combine to form fibres. Tendon fibroblasts reside between collagen fibres. Fibres are surrounded by a connective tissue, the endotenon, which also contains fibroblasts. Fibres combine to form fascicles. Tendons are ensheathed by an outer layer of connective tissue, the epitenon, which is surrounded by another layer of connective tissue, the paratenon. Together, the epitenon and paratenon external sheaths compose the peritenon. **c** | Electron microscopy of a transverse section of adult mouse tail tendon. **d** | Histological staining of a transverse section of a mouse tail tendon fibre showing the nuclei of tendon cells (purple) and collagen (pink). **e** | Longitudinal section of adult mouse Achilles tendon stained with anti-type I collagen antibody (green) and Hoechst to visualize the nuclei of tendon cells (blue).

risk factors for tendon disorders include sex, age and diseases such as type 2 diabetes mellitus and obesity.^{6,34} Genetic predisposition might also influence risk of tendon injuries.³⁵ The main recognized extrinsic factor for tendon injury is abnormal loading on tendons, which is linked to physiological exercise, sport and specific work settings. Tendinopathy is thought to result from repetitive abnormal mechanical loading, whereas acute tendon injury often results after one isolated overloading

event.^{1-3,5,10,28,36} Aberrant mechanical stimulation induces the production of biological factors, including metalloproteinases, growth factors and prostaglandins, which can all lead to ECM remodelling defects.^{10,37} Moreover, excessive mechanical loading has been proposed to cause aberrant differentiation of TSCs into non-tendon cells.³⁸

Treatment of tendon injuries

First-line therapeutic options differ for chronic and acute tendon injuries. The primary goal of tendinopathy treatment is to reduce pain, mainly through the use of topical or systemic anti-inflammatory drugs, whereas surgical techniques aim to repair ruptured tendons.^{1,39,40} The outcomes of reconstructive surgery differ depending on the type and location of the injury.¹ Independently of surgical procedures or nonsurgical management of tendon injury, rerupture often occurs because scarring results in weakened tendon tissue.^{1,2,36} For both chronic and acute tendon injuries, exercise-based rehabilitation is indicated.⁴¹ Eccentric exercise therapy (involving active lengthening of muscle and tendon) has become the treatment of choice and is considered as the most efficient for tendinopathy.^{3,39,40} Mechanical loading is also integrated into clinical postoperative rehabilitation protocols.³⁶ Active movements are beneficial for flexor tendon healing.⁴¹ The use of autologous growth factors is another therapeutic approach that is gaining in popularity for the treatment of tendon injury. Platelet-rich plasma (PRP) is a blood derivative containing high levels of growth factors, known to promote tissue healing.⁴² Because PRP is readily available and autologous PRP therapy is considered safe, PRP has been introduced into clinical therapy for tendinopathy and acute tendon injury. However, the benefits of PRP injection for tendon recovery remain controversial.^{43,44} Extracorporeal shock-wave therapy has demonstrated some efficacy but only in calcified tendinitis of the shoulder; less-conventional procedures, such as phonophoresis, therapeutic ultrasonography or low-level laser therapy, are other options for the treatment of tendon injury.^{3,40} Stem-cell-based therapy is also an attractive option, which could become available in the future.¹

Surgery, specific exercise-based therapy and autologous growth factor injection are the main current treatments for tendon injuries. In addition to being moderately effective or controversial, the underlying mechanisms of these treatments are not fully understood. More specifically, the healing potential of tendon after injury needs to be further elucidated.

Experimental models of tendon injury

Animal models offer an attractive framework for investigating the molecular and cellular mechanisms underlying tendon injury, and have been extensively developed for this purpose.^{2,45} Because the pathogenesis of tendinopathy is unclear, animal models attempting to reproduce the pathology (mainly by inducing mechanical or chemical injuries) do not perfectly mimic the disease.⁴⁵⁻⁴⁸ However, animal models have been used extensively to study tendon healing following total or partial tendon transection.^{2,45}

Table 1 | Mechanical and biological factors in tendon development, homeostasis and repair

Factor	Effects on tendon development, homeostasis and healing/repair <i>in vivo</i> *	Effects on stem cells [†] or engineered tendon [§] tissue <i>in vitro</i>	Effects on tendon repair in animal models of injury
Mechanical factors			
Overloading	Promotes homeostasis ^{38,121,123} Promotes repair at the midtendon ^{128–130} but inhibits repair at the enthesis ¹²⁷	Promotes tenogenesis in 2D culture ^{38,126} Promotes tendon construct formation in 3D culture ^{106,125}	ND
Underloading	No tendon formation ^{102,117–119} Negative effect on homeostasis ^{119,120} Inhibits repair at the midtendon ^{128–130} but promotes repair at the enthesis ¹²⁶	Does not promote or inhibit tendon construct formation in 3D culture ⁶⁸	ND
TGF-β-SMAD2/3 signalling pathway			
TGF- β 1, TGF- β 2, TGF- β 3	In development, knocking out <i>Tgfb2</i> and <i>Tgfb3</i> results in loss of fetal tendons ²⁰	Promotes tenogenesis in 2D culture ^{10,27,74–76} In 3D culture, enhances the molecular, histological and mechanical properties of engineered tendon constructs ^{68,69,77–79}	Local delivery of TGF- β improves molecular, histological and mechanical properties of healed tendon ^{49,81}
GDF-8 (myostatin)	Altered fetal myogenesis in <i>Mst</i> ^{-/-} mice ¹³⁵ Tendons of <i>Mst</i> ^{-/-} mice are hypocellular ⁸²	Promotes tenogenic differentiation of stem cells in 2D culture ⁸²	ND
SMAD3	During development, <i>Smad3</i> ^{-/-} mice have reduced gene and protein expression of matrix components in tendon ⁸⁴ <i>Smad3</i> ^{-/-} mice have reduced expression of tendon genes in tendon ⁸⁴ <i>Smad3</i> ^{-/-} mice have defective healing in tendon after injury ⁵⁷	ND	ND
BMP signalling pathway (SMAD1/5/8)			
BMP-4	<i>Bmp4</i> ^{-/-} (using Prx1-Cre as a deleter) results in enthesitis defects ⁹⁹	ND	ND
GDF-5 (BMP-14), GDF-6 (BMP-13), GDF-7 (BMP-12)	In development, knocking out <i>Gdf7</i> results in defects in neuronal and seminal vesicle identity; ^{94,95} knocking out <i>Gdf6</i> or <i>Gdf5</i> results in joint and skeletal defects ^{92,93} <i>Gdf7</i> ^{-/-} mice have a mildly altered tendon phenotype; ⁹⁷ knocking out <i>Gdf6</i> or <i>Gdf5</i> disrupts tail and Achilles tendon phenotype; ^{96,98} knocking out <i>Gdf5</i> delays tendon healing after injury ⁵⁶	Increases expression of tendon and cartilage markers in 2D stem cell cultures ^{85,86,101} In 3D culture, enhances the molecular properties of tendon constructs ⁸⁸	Improves molecular, histological and mechanical properties of healed tendon (increase in tendon and cartilage markers) ^{51,85,89}
BMP-2/SMAD8	ND	Promotes tenogenesis in 2D culture ⁸⁷	Improves histological and mechanical properties of healed tendon ⁹⁰
FGF signalling pathway			
FGF-2 (basic FGF)	ND	In 2D culture, promotes tenogenesis of human ¹³⁶ and rat ¹³⁷ stem cells, but does not promote tenogenesis in canine stem cells ¹³⁸ In 3D culture, enhances the molecular, histological and mechanical properties of tendon constructs (rabbit) ¹³⁹	Improves molecular, and mechanical properties of healed tendon in chick model ^{108,109} but not in canine model ¹⁰⁷
FGF-4	Sufficient for Scx expression in chick embryos ^{102–104} Inhibits Scx expression in mouse embryos ⁷³	Does not promote tenogenesis in mouse stem cells ^{73,74}	ND
ERK MAPK	Blockade of ERK MAPK increases Scx expression in mouse embryo cultures ⁷³	Does not promote tenogenesis in mouse stem cells ⁷³ In 3D culture, enhances the molecular and mechanical properties of tendon constructs (human) ¹⁰⁶	ND
Other signalling pathways			
PDGF-BB	Knocking out <i>Pdgfb</i> affects vascular phenotype (pericytes) ⁶¹ Embryonic lethal ⁶¹	In 3D culture, increases cell proliferation ^{140,141}	Improves histological and mechanical properties of healed tendon ^{49,62}
VEGF	Knocking out <i>Vegf</i> affects vascular phenotype (endothelial cells) ⁶⁰	ND	Does not improve mechanical properties of healed tendon ⁴⁹
IGF-1	Knocking out <i>Igf1</i> results in postnatal growth defects ⁶⁵	In 3D culture, enhances molecular, histological and mechanical properties of tendon constructs ^{68,69}	Improves histological and mechanical properties of healed tendon ^{49,70}

Table 1 (Cont.) | Mechanical and biological factors in tendon development, homeostasis and repair

Factor	Effects on tendon development, homeostasis and healing/repair <i>in vivo</i> *	Effects on stem cells [†] or engineered tendon [§] tissue <i>in vitro</i>	Effects on tendon repair in animal models of injury
Transcription factors			
Scleraxis	Knocking out <i>Scx</i> results in fetal tendon defects ²⁰ and adult tendon defects ²⁰	Promotes tenogenesis in 2D stem cell culture ¹¹² In 3D culture, enhances molecular properties of tendon constructs ¹¹³	Improves molecular, histological and mechanical properties of healed tendon ^{113,114}
Mohawk	Knocking out <i>Mkx</i> results in decreased <i>Col1a1</i> transcription in late-stage fetal development ²⁴ and decreased type I collagen in mature tendon ²⁵	Promotes tenogenesis in 2D stem cell culture ^{115,116} In 3D culture, enhances molecular and histological properties of tendon constructs ¹¹⁵	Improves molecular, histological and mechanical properties of healed tendon ^{115,116}
EGR1	Knocking out <i>Egr1</i> results in decreased <i>Col1a1</i> transcription in late-stage fetal development ¹⁸ and decreased expression of <i>Col1a1</i> , <i>Scx</i> and <i>Tnmd</i> in mature tendon ²⁷ Knocking out <i>Egr1</i> results in decreased expression of tendon genes after injury ²⁷	Promotes tenogenesis in 2D stem cell culture ²⁷ In 3D culture, enhances molecular and histological properties of tendon constructs ²⁷	Improves molecular and histological properties of healed tendon ²⁷
*Tendon phenotypes during development and homeostasis have been reported mainly from mouse genetic work; tendon healing has been studied in three strains of genetically engineered mice. [†]Tenogenic differentiation of stem cells in 2D cultures as defined by ectopic activation of <i>Scx</i>, <i>Col1a1</i> or <i>Tnmd</i>. [§]Engineered tendons made of 3D-cultured stem cells; effects assessed using molecular, histological and mechanical criteria. Factor delivered ectopically in an animal model of tendon injury; effects on repair assessed using molecular, histological and mechanical criteria. Abbreviations: BMP, bone morphogenetic protein; EGR1, early growth response protein 1; ERK, extracellular signal-regulated kinase; FGF, fibroblast growth factor; GDF, growth/differentiation factor; IGF-1, insulin-like growth factor 1; MAPK, mitogen-activated protein kinase; ND, not determined; PDGF-BB, platelet-derived growth factor BB; TGF, transforming growth factor; VEGF, vascular endothelial growth factor.			

Studies using these tendon transection models have provided insights into the molecular processes of tendon repair, mainly by identifying growth factors. These growth factors have attracted much attention for their potential use in treating tendon disorders.^{1,49}

Sequential phases of tendon repair

Tendon repair after injury involves the sequential and overlapping phases of inflammation, cell proliferation, cell migration and remodelling.^{1,2} These successive phases ultimately result in the production and spatial organization of type I collagen. However, healed tendons do not regain the chemical and mechanical properties of native uninjured tendons. Because the repair process induces the formation of scar tissue and not native tendon tissue, the tensile strength of healed tendon can be one-third that of intact tendon in human.³¹

The origin or source of cells involved in tendon repair remains undefined; they could arise from, for example, blood, fat or tendon sheaths. Moreover, the contribution of TSCs¹⁴ to the tendon healing process is debated.^{16,50} As mentioned above, TSCs that could contribute to tendon repair have been identified in the tendon proper (including endotenon) and in the external tendon sheaths (epitenon and paratenon).¹⁶ However, the molecular mechanisms controlling the proliferation, migration and differentiation of TSCs during tendon repair are not well understood.

Molecular response to tendon injury

After injury, a large panel of growth factors and cytokines are released by the injured tendons and adjacent tissues, including interleukins, TNF, vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF), FGF, TGF- β , connective tissue growth factor (CTGF), epidermal growth factor (EGF) and insulin-like growth factor (IGF)-1, among others.^{51,52} The early inflammatory

phase (immediately after tendon trauma) is associated with the release of interleukins and TNF produced by proinflammatory M1 macrophages, whereas the secondary inflammatory response involves anti-inflammatory M2 macrophages, which produce growth factors involved in neovascularization, such as VEGF, FGF and PDGF, and profibrotic factors, such as TGF- β and CTGF.⁵³ In addition to growth factors, tendon injury leads to a massive increase in the relative mRNA levels of genes encoding collagen (*Col1a1*, *Col1a2*, *Col3a1*, *Col12a1*, *Col14a1*) and tendon-associated molecules including tenomodulin (*Tnmd*), tenascin (*Tnc*) and proteoglycans in animal models.^{27,51,54} The ECM-gene upregulation observed in animal models of tendon injury is also observed in human tendinopathy.³³ The expression of genes encoding the tendon-associated transcription factors scleraxis (*Scx*), Mohawk (*Mkx*) and EGR1 (*Egr1*) is also upregulated after tendon injury.^{27,51,54,55} The timing of production, and the precise cellular origin, of all these molecules is not completely established. However, fibroblasts from external tendon sheaths are widely believed to produce cytokines and growth factors, whereas fibroblasts from tendon proper are thought to produce type I collagen and ECM components.^{16,50} Although gene expression and cytokine release is massively increased after tendon injury, only three factors have been shown, on the basis of genetic loss-of-function experiments, to be required for a complete tendon repair response: SMAD3, growth/differentiation factor (GDF)-5 and EGR1 (Table 1).^{27,56,57}

Growth factor effects in experimental models

Most of the growth factors and cytokines activated during tendon healing are involved in the inflammatory response, making it difficult to deduce which pathways are involved in a protenogenic effect. Because growth factor and cytokine therapies seem promising for tendon

repair, the tendon-promoting effect of these growth factors has been extensively studied in laboratories *in vitro*, using stem cell cultures, and *in vivo*, in tendon injury models.

In 2D stem cell culture systems, the ectopic application of a growth factor can test the ability of the growth factor to induce tenocyte differentiation. However, full tenogenesis is difficult to assess in 2D-culture systems. To date, the best marker of tenocyte differentiation is the type II transmembrane glycoprotein tenomodulin.^{23,58,59} 3D culture systems have been developed to engineer tendons from MSCs. In these 3D systems, the ectopic application of growth factors can test the ability of the growth factor to promote (rather than induce) tendon formation.

In vivo, the tendon-promoting effects of growth factors have been extensively tested in animal models, by ectopic delivery of the growth factors following tendon injury. However, overexpression of a factor that is already upregulated following injury might not be the best approach to test its tendon-promoting effect. Several methods have been used to deliver growth factors to injured tendons, mainly local injection of recombinant proteins, with or without the use of biomaterial carriers, and of stem cells transduced with growth factors. Although compelling arguments are made for the preferred protocol in each study, no consensus exists on the best method to employ for *in vivo* growth-factor delivery.¹

Insights from developmental biology

Adult tissue regeneration is generally thought to recapitulate developmental processes. Consequently, knowledge of the molecular mechanisms underlying tendon development should clarify tendon repair processes. Consistent with their well-established role in vasculogenesis during development,^{60,61} the ectopic application of VEGF and PDGF has been shown to increase cell proliferation and to promote angiogenesis in injured tendons.^{49,62} However, ectopic VEGF delivery does not increase *TNMD* expression⁶³ and has deleterious effects on tendon repair in animal models.⁴⁹ Recombinant human PDGF-BB is considered, on the basis of morphological and mechanical criteria, to promote tendon repair in animal models,⁶² but it is also pivotal for bone repair.⁶⁴ We believe that PDGF-BB promotes tissue repair via its generic angiogenic, chemotactic and mitogenic properties.⁶² Mice lacking Igf-1 (*Igf1*^{-/-}) have severe postnatal general growth defects,⁶⁵ consistent with a general anabolic function of this hormone. IGF-1 also has an anabolic effect on human and mouse tendons^{66,67} and promotes collagen synthesis in engineered human tendons.^{68,69} IGF-1 has been reported to improve not only tendon repair,^{49,70} but also articular cartilage repair.⁷¹ The activation of scleraxis or tenomodulin upon application of VEGF, PDGF and IGF-1 has never been reported in stem cells other than TSCs,⁴⁹ indicating that these growth factors do not have the ability to induce tenogenesis in stem cells (Table 1). The outcome of PDGF and IGF-1 application for tendon repair seems to be beneficial but is not tendon-specific.

Embryological experiments and genetic analyses have identified TGF- β and FGF as the main growth factors involved in vertebrate tendon development.^{30,72} In addition, bioinformatics analysis of the transcriptome of mouse-limb tendon cells also identified the TGF- β -SMAD2/3 and FGF-ERK/MAPK signalling pathways as the two pathways most strongly modified during mouse tendon development.⁷³

TGF- β ligands

Consistent with the requirement and sufficiency of a TGF- β signal for mouse tendon development,^{19,20,73,74} ectopic delivery of a TGF- β ligand systematically promotes the tenogenic differentiation of stem cells in 2D-culture,^{20,27,75,76} the formation of 3D-engineered tendons,^{68,77–80} and the tendon repair response^{49,81} (Table 1). Genetic analysis in mice has shown that TGF- β 2 and TGF- β 3 are the TGF- β ligands required for tendon fetal development, since fetal tendons are lost in *Tgfb2*^{-/-} and *Tgfb3*^{-/-} mice.²⁰ In tendon homeostasis and repair, other TGF- β ligands, such as growth/differentiation factor 8 (GDF-8, encoded by *MSTN*), could also be involved, as *Mstn*^{-/-} mice have hypocellular tendons, in addition to hypertrophic muscles.⁸² During tendon development and repair, the TGF- β -related ligands seem to have consistent tendon-promoting effects (Table 1). These TGF- β ligands transduce biological responses mainly via the intracellular SMAD2/3 pathway.⁸³ Consistently, intracellular SMAD3 has been shown to be involved in tendon development and repair.^{57,73,84}

BMP-related ligands

The bone morphogenetic protein (BMP)-related members of the TGF- β superfamily also have tendon-promoting effects in 2D and 3D cultures of stem cells^{85–88} and during the tendon repair response^{51,85,89,90} (Table 1). BMP ligands transduce biological responses via the intracellular SMAD1/5/8 pathway and are well-known to be involved in bone formation and repair.^{83,91} In mouse embryos, *Gdf5* and *Gdf6* are not obviously involved in tendon development but rather in joint and cartilage development.^{92,93} *Gdf7*^{-/-} mice have neuronal and seminal vesicle defects.^{94,95} Only subtle tendon defects have been reported in tail or Achilles tendons of *Gdf5*^{-/-}, *Gdf6*^{-/-} and *Gdf7*^{-/-} adult mutant mice,^{96–98} which could be the consequences of developmental defects in the skeleton and joints. We believe that the tendon-promoting effects observed after ectopic application of BMP-related ligands using *in vitro* and *in vivo* systems (Table 1) is related to enthesis development. Enthesis formation involves different cellular and molecular processes than those involved in the development of tendon proper. Enteses are derived from a unique set of progenitor cells that express both *Scx* and *Sox9*, which are specified independently of cartilage progenitors.^{99,100} BMP signalling has been shown to be involved in enthesis development in the mouse embryo.⁹⁹ Consistent with this finding, application of ectopic BMP-related ligands to stem cells often leads to the upregulation of cartilage and bone markers, in addition to tendon

markers (Table 1).^{51,86,89,101} One plausible hypothesis for this observation is that the ectopic application of BMP-related ligands promotes enthesis formation in animal models of tendon injury.

FGF-ERK/MAPK signalling

The FGF-ERK/MAPK (mitogen-activated protein kinase) signalling pathway has been shown to be involved in vertebrate tendon development, although the involvement of FGF differs in chick and mice embryos. FGF-ERK/MAPK signalling has been shown to be sufficient and required for tendon formation in chick embryos.^{102–105} FGF does not have similar functions in mouse tendon formation, as blockade of ERK/MAPK signalling has been shown to promote *Scx* expression in mouse tendon progenitors and stem cells.^{23,73,74} However, in 3D-engineered tendons made of human tendon cells, ERK/MAPK activity and collagen content are directly correlated.¹⁰⁶ This evidence suggests that the effects of FGF-ERK/MAPK signalling vary either at different steps of tenogenesis or between species. *In vivo*, the outcome for tendon repair of ectopic FGF delivery after tendon injury is controversial. Exogenous basic FGF consistently increases cell proliferation in the early stages of tendon repair,⁴⁹ but diminishes the mechanical strength of injured tendons.¹⁰⁹ Interestingly, consistently with the positive effect of FGF in chick tendon development, the ectopic application of FGF has a beneficial effect on tendon repair in a chick flexor injury model, on the basis of *Scx* expression and improved tendon strength.^{108,109}

Combination of growth factors

In addition to ectopic delivery of a single growth factor, platelet-rich plasma (PRP) has also been used based on the hypothesis that the delivery of several molecules will boost the tendon repair process. However, studies in animal models have not conclusively shown that PRP affects tendon repair, which is consistent with the contradictory outcomes of the clinical use of PRP for managing tendon injury.^{43,44,110} The different results could be explained by variability in the PRP preparations between studies.^{43,44,111} Moreover, the application of a combination of growth factors and cytokines will simultaneously activate multiple intracellular signalling pathways, which do not necessarily all induce a tendon-promoting effect.

Transcription factors

All three transcription factors associated with tendon development and *Col1a1* transcription are activated following tendon injury.^{27,51,54,55} Scleraxis, Mohawk and EGR1 have been shown to trigger tendon differentiation in stem cells (indicated by tenomodulin expression), to promote the formation of 3D-engineered tendons and to improve repair in animal models of tendon injury (Table 1).^{27,112–116} The genetic link between these three factors is not completely understood, but *Mkx* and *Scx* seem to act in independent genetic regulatory cascades in mouse tendons.^{24,25} However, ectopic delivery of Mohawk activates *Scx* expression in mouse stem cells¹¹⁵ but not in human stem cells.¹¹⁶

Tendon development versus tendon repair

The molecular and cellular processes that regulate tendon development and tendon repair, although not completely identified, have some similarities. The TGF- β -SMAD2/3 and FGF-ERK/MAPK signalling pathways are both involved in both tendon development and tendon repair (Table 1). Interestingly, molecular profiling in human samples of tendinopathy also revealed misregulation of components of TGF- β signalling pathways.³³ We suspect that less-studied (in the context of tendon biology) pathways, such as Wnt and calcium pathways, are also important for tendon repair as they have been identified as being modulated in mouse tendon development and in human tendinopathy in studies using global approaches.^{33,73} A systematic study of how signalling pathways integrate and interact with each other during tendon development and repair is needed.

Tendon mechanobiology

Development and homeostasis

A major parameter in tendon biology is mechanical stimulation. During development, the absence of muscle and, consequently, of movement, impairs tendon formation.^{105,117–119} The initiation of limb tendon development is independent of mechanical parameters as tendon development is initiated in muscleless limbs. However, tendon development is later arrested in muscleless limbs, indicating that mechanical forces are required for complete tendon formation.^{105,117–119}

Mechanical forces are also critical for adult tendon homeostasis in animal models and in humans.^{21,28} A loss of continuous force transmission from skeletal muscles leads to a reduction of tendon size and impaired tendon biomechanical properties in animal models.²⁸ This mechanical change also drastically diminishes the expression of tendon markers, including scleraxis, and changes ECM composition in mice.¹²⁰ Conversely, conditions of increased mechanical loading increase the synthesis of collagen and other ECM components,¹²¹ whereas conditions of decreased loading lead to a progressive decrease in rates of tendon collagen synthesis in human tendons.¹²² Treadmill training increases *Col1a1*, *Scx* and *Tnmd* expression levels in mouse tendons.^{38,123} Moreover, *Tnmd* expression levels in Achilles and patellar tendons are positively correlated with the intensity of the treadmill running.³⁸ *SCX* and *COL1A1* expression is also enhanced in 3D-engineered tendons when subjected to mechanical stimulation.^{124,125} Increased expression of *SCX* and *COL1A1* is also positively correlated with the strain of cyclic loading in 3D-bioartificial tendons.¹²⁵ Tendon cells sense and respond to changes in their mechanical environment. Consequently, even in 2D cell cultures, cyclic mechanical stretching increases *Scx* and *Col1a1* transcript levels.^{38,126}

Thus, on the basis of collagen synthesis and tendon gene expression a physiological increase of mechanical load is generally beneficial, whereas a diminution of mechanical load is detrimental, for tendon formation during tendon development and homeostasis.

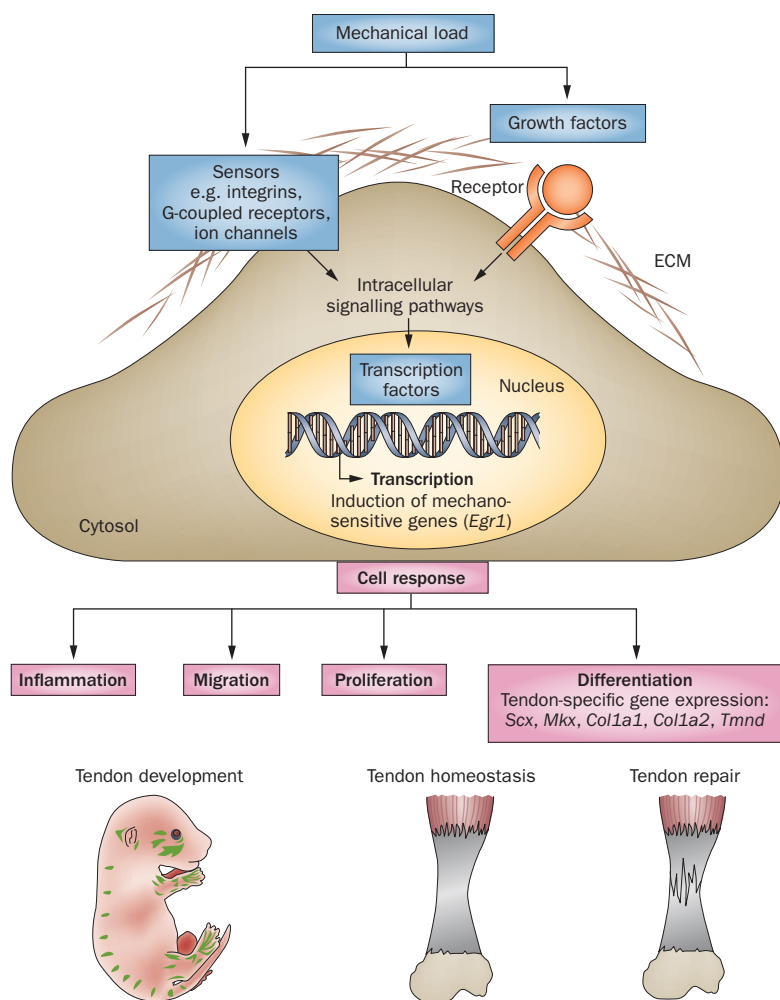


Figure 2 | Tendon mechanobiology. Mechanical stimuli are transduced via the ECM, growth factors, receptors, intracellular pathways and transcription factors to induce specific responses in tendon cells. EGR1 is one mechanosensitive transcription factor involved in the tendon cell response. Mechanical and biological parameters converge to drive tendon gene transcription. Similarities exist in the mechanotransduction processes of tendon development, homeostasis and repair. Abbreviations: ECM, extracellular matrix; EGR1, early growth response protein 1.

Mechanical loading during tendon repair

Mechanical forces are also involved in the repair process after tendon injury.¹²⁷ Studies in rats established that mechanical loading stimulates tendon healing, whereas immobilization is detrimental to the healing process, in injured Achilles tendons.^{128–130} However, the beneficial effect of mechanical loading depends on the site of injury and tendon type, as immobilization in a cast improved enthesis healing in rotator cuff tendons.¹³¹ Thus, the influence of mechanical loading on tendon repair is not the same for different tendon types and tendon regions.¹²⁷ The differences in outcomes probably arise from variation in the mechanical demands placed on different tendon areas (for example, mid-tendon versus enthesis) and also on different tendon types.

Molecular transduction of mechanical load

Although the importance of mechanical forces for tendon biology is recognized, the conversion of mechanical

stimuli into a biochemical response is not fully understood. Mechanical alterations in cells result in changes to ECM organization, cytoskeletal organization and gene transcription.¹² We still need to understand how mechanical forces are converted to molecular and cellular processes during tendon development, homeostasis, ageing and repair. To date, few studies have shown a causal relationship between mechanical forces, gene expression, growth factor production and increased collagen synthesis in tendon. However, the two signalling pathways involved in tendon development, TGF- β -SMAD2/3 and FGF-ERK/MAPK, are known to transduce biological responses upon mechanical stress.¹² Moreover, mechanical forces have been shown to induce *Scx* expression through the activation of SMAD2/3-mediated TGF- β signalling in adult mouse tendons.¹²⁰ Stretch experiments on *in vitro* tendon constructs led to an increase in ERK/MAPK phosphorylation.¹⁰⁶ Moreover, the transcription factors EGR1 and EGR2, which are involved in chick and mouse tendon development,¹⁸ (Table 1) are also encoded by mechanosensitive genes.¹³² *Egr1* and *Egr2* gene expression has been reported to be increased 15 min after a loading episode in injured rat Achilles tendons.^{133,134} This observation leads to the interesting hypothesis that the EGR1 and EGR2 transcription factors could be molecular sensors of mechanical parameters driving the tendon differentiation process during tendon repair and development. Moreover, *Egr1* positively regulates *Tgfb2* transcription by direct recruitment to the regulatory regions of the *Tgfb2* gene in normal and injured adult mouse tendons.²⁷ Interestingly, Mohawk also positively regulates *Scx* expression via direct activation of *Tgfb2*.¹¹⁵ These observations lead to the hypothesis that forces could induce the transcription of molecular sensors that will drive the production of growth factors promoting tendon differentiation (Figure 2). We believe that the mechanotransduction processes driving tendon cell proliferation and differentiation during tendon development, homeostasis and repair have similarities (Figure 2).

Conclusions

A wealth of information is available regarding tendon physiology and pathologies (diagnosis, treatments) and for tendon healing, repair and development (mostly from experimental models), which needs to be synthesized. Although basic science cannot substitute for clinical research, we believe that the former is fundamental to understanding tendon pathologies. Mechanical loading is important for tendon development, tendon homeostasis, tendon repair and clinical rehabilitation protocols for tendon injuries. Mechanobiology involves a combination of mechanical and biological parameters that need to be hierarchized in the context of healthy and pathological tendons. Deciphering how these mechanotransduction pathways affect transcriptional activity and cellular responses in the context of tendon biology is prerequisite to the design of novel and relevant therapies for tendon injury.

1. Docheva, D., Muller, S. A., Majewski, M. & Evans, C. H. Biologics for tendon repair. *Adv. Drug Deliv. Rev.* <http://dx.doi.org/10.1016/j.addr.2014.11.015> (2014).
2. Voleti, P. B., Buckley, M. R. & Soslowsky, L. J. Tendon healing: repair and regeneration. *Annu. Rev. Biomed. Eng.* **14**, 47–71 (2012).
3. Kaux, J. F., Forthomme, B., Goff, C. L., Crielaard, J. M. & Croisier, J. L. Current opinions on tendinopathy. *J. Sports Sci. Med.* **10**, 238–253 (2011).
4. Maffulli, N., Khan, K. M. & Puddu, G. Overuse tendon conditions: time to change a confusing terminology. *Arthroscopy* **14**, 840–843 (1998).
5. Magnusson, S. P., Langberg, H. & Kjaer, M. The pathogenesis of tendinopathy: balancing the response to loading. *Nat. Rev. Rheumatol.* **6**, 262–268 (2010).
6. Magnan, B., Bondi, M., Pierantoni, S. & Samaila, E. The pathogenesis of Achilles tendinopathy: a systematic review. *Foot Ankle Surg.* **20**, 154–159 (2014).
7. Rees, J. D., Stride, M. & Scott, A. Tendons—time to revisit inflammation. *Br. J. Sports Med.* **48**, 1553–1557 (2014).
8. Kannus, P. & Jozsa, L. Histopathological changes preceding spontaneous rupture of a tendon. A controlled study of 891 patients. *J. Bone Joint Surg. Am.* **73**, 1507–1525 (1991).
9. American Academy of Orthopaedic Surgeons. Rotator cuff tears, *OrthoInfo* [online], <http://orthoinfo.aaos.org/topic.cfm?topic=A00064> (2011).
10. Thornton, G. M. & Hart, D. A. The interface of mechanical loading and biological variables as they pertain to the development of tendinosis. *J. Musculoskelet. Neuronal Interact.* **11**, 94–105 (2011).
11. Sharma, P. & Maffulli, N. Tendon injury and tendinopathy: healing and repair. *J. Bone Joint Surg. Am.* **87**, 187–202 (2005).
12. Humphrey, J. D., Dufresne, E. R. & Schwartz, M. A. Mechanotransduction and extracellular matrix homeostasis. *Nat. Rev. Mol. Cell Biol.* **15**, 802–812 (2014).
13. Mienaltowski, M. J. & Birk, D. E. Structure, physiology, and biochemistry of collagens. *Adv. Exp. Med. Biol.* **802**, 5–29 (2014).
14. Bi, Y. et al. Identification of tendon stem/progenitor cells and the role of the extracellular matrix in their niche. *Nat. Med.* **13**, 1219–1227 (2007).
15. Zhang, J. & Wang, J. H. Characterization of differential properties of rabbit tendon stem cells and tenocytes. *BMC Musculoskelet. Disord.* **11**, 10 (2010).
16. Mienaltowski, M. J., Adams, S. M. & Birk, D. E. Regional differences in stem cell/progenitor cell populations from the mouse achilles tendon. *Tissue Eng. Part A* **19**, 199–210 (2013).
17. Zelzer, E., Blitz, E., Killian, M. L. & Thomopoulos, S. Tendon-to-bone attachment: from development to maturity. *Birth Defects Res. C Embryo Today* **102**, 101–112 (2014).
18. Lejard, V. et al. EGR1 and EGR2 involvement in vertebrate tendon differentiation. *J. Biol. Chem.* **286**, 5855–5867 (2011).
19. Lorde-Diez, C. I., Montero, J. A., Martinez-Cue, C., Garcia-Porrero, J. A. & Hurler, J. M. Transforming growth factors β coordinate cartilage and tendon differentiation in the developing limb mesenchyme. *J. Biol. Chem.* **284**, 29988–29996 (2009).
20. Pryce, B. A. et al. Recruitment and maintenance of tendon progenitors by TGF β signaling are essential for tendon formation. *Development* **136**, 1351–1361 (2009).
21. Heinemeier, K. M. & Kjaer, M. *In vivo* investigation of tendon responses to mechanical loading. *J. Musculoskelet. Neuronal Interact.* **11**, 115–123 (2011).
22. Yun, Y. R. et al. Fibroblast growth factors: biology, function, and application for tissue regeneration. *J. Tissue Eng.* **2010**, 218142 (2010).
23. Murchison, N. D. et al. Regulation of tendon differentiation by scleraxis distinguishes force-transmitting tendons from muscle-anchoring tendons. *Development* **134**, 2697–2708 (2007).
24. Liu, W. et al. The atypical homeodomain transcription factor Mohawk controls tendon morphogenesis. *Mol. Cell Biol.* **30**, 4797–4807 (2010).
25. Ito, Y. et al. The Mohawk homeobox gene is a critical regulator of tendon differentiation. *Proc. Natl Acad. Sci. USA* **107**, 10538–10542 (2010).
26. Lejard, V. et al. Scleraxis and NFATc regulate the expression of the pro- α 1(I) collagen gene in tendon fibroblasts. *J. Biol. Chem.* **282**, 17665–17675 (2007).
27. Guerquin, M. J. et al. Transcription factor EGR1 directs tendon differentiation and promotes tendon repair. *J. Clin. Invest.* **123**, 3564–3576 (2013).
28. Wang, J. H., Guo, Q. & Li, B. Tendon biomechanics and mechanobiology—a minireview of basic concepts and recent advancements. *J. Hand Ther.* **25**, 133–140 (2012).
29. Shwartz, Y., Blitz, E. & Zelzer, E. One load to rule them all: mechanical control of the musculoskeletal system in development and aging. *Differentiation* **86**, 104–111 (2013).
30. Tozer, S. & Duprez, D. Tendon and ligament: development, repair and disease. *Birth Defects Res. C Embryo Today* **75**, 226–236 (2005).
31. Leadbetter, W. B. Cell-matrix response in tendon injury. *Clin. Sports Med.* **11**, 533–578 (1992).
32. Xu, Y. & Murrell, G. A. The basic science of tendinopathy. *Clin. Orthop. Relat. Res.* **466**, 1528–1538 (2008).
33. Jelinsky, S. A. et al. Regulation of gene expression in human tendinopathy. *BMC Musculoskelet. Disord.* **12**, 86 (2011).
34. Nourissat, G., Houard, X., Sellam, J., Duprez, D. & Berenbaum, F. Use of autologous growth factors in aging tendon and chronic tendinopathy. *Front. Biosci. (Elite Ed.)* **E5**, 911–921 (2013).
35. Ribbans, W. J. & Collins, M. Pathology of the tendo Achillis: do our genes contribute? *Bone Joint J.* **95-B**, 305–313 (2013).
36. Freedman, B. R., Gordon, J. A. & Soslowsky, L. J. The Achilles tendon: fundamental properties and mechanisms governing healing. *Muscles Ligaments Tendons J.* **4**, 245–255 (2014).
37. Davis, M. E., Gumucio, J. P., Sugg, K. B., Bedi, A. & Mendias, C. L. MMP inhibition as a potential method to augment the healing of skeletal muscle and tendon extracellular matrix. *J. Appl. Physiol.* (1985). **115**, 884–891 (2013).
38. Zhang, J. & Wang, J. H. The effects of mechanical loading on tendons—an *in vivo* and *in vitro* model study. *PLoS ONE* **8**, e71740 (2013).
39. Rees, J. D., Wilson, A. M. & Wolman, R. L. Current concepts in the management of tendon disorders. *Rheumatology (Oxford)* **45**, 508–521 (2006).
40. Childress, M. A. & Beutler, A. Management of chronic tendon injuries. *Am. Fam. Physician* **87**, 486–490 (2013).
41. Khan, K. M. & Scott, A. Mechanotherapy: how physical therapists' prescription of exercise promotes tissue repair. *Br. J. Sports Med.* **43**, 247–252 (2009).
42. Jeong, D. U. et al. Clinical applications of platelet-rich plasma in patellar tendinopathy. *Biomed. Res. Int.* **2014**, 249498 (2014).
43. Wang, J. H. Can PRP effectively treat injured tendons? *Muscles Ligaments Tendons J.* **4**, 35–37 (2014).
44. Guevara-Alvarez, A., Schmitt, A., Russell, R. P., Imhoff, A. B. & Buchmann, S. Growth factor delivery vehicles for tendon injuries: mesenchymal stem cells and platelet rich plasma. *Muscles Ligaments Tendons J.* **4**, 378–385 (2014).
45. Hast, M. W., Zuskov, A. & Soslowsky, L. J. The role of animal models in tendon research. *Bone Joint Res.* **3**, 193–202 (2014).
46. Lui, P. P., Maffulli, N., Rolf, C. & Smith, R. K. What are the validated animal models for tendinopathy? *Scand. J. Med. Sci. Sports* **21**, 3–17 (2011).
47. Dirks, R. C. & Warden, S. J. Models for the study of tendinopathy. *J. Musculoskelet. Neuronal Interact.* **11**, 141–149 (2011).
48. Heinemeier, K. M. et al. Uphill running improves rat Achilles tendon tissue mechanical properties and alters gene expression without inducing pathological changes. *J. Appl. Physiol.* (1985). **113**, 827–836 (2012).
49. Halper, J. Advances in the use of growth factors for treatment of disorders of soft tissues. *Adv. Exp. Med. Biol.* **802**, 59–76 (2014).
50. Dymant, N. A. et al. The paratenon contributes to scleraxis-expressing cells during patellar tendon healing. *PLoS One* **8**, e59944 (2013).
51. Jelinsky, S. A. et al. Treatment with rhBMP12 or rhBMP13 increase the rate and the quality of rat Achilles tendon repair. *J. Orthop. Res.* **29**, 1604–1612 (2011).
52. Manning, C. N. et al. The early inflammatory response after flexor tendon healing: a gene expression and histological analysis. *J. Orthop. Res.* **32**, 645–652 (2014).
53. Chazaud, B. Macrophages: supportive cells for tissue repair and regeneration. *Immunobiology* **219**, 172–178 (2014).
54. Scott, A., Sampaio, A., Abraham, T., Duronio, C. & Underhill, T. M. Scleraxis expression is coordinately regulated in a murine model of patellar tendon injury. *J. Orthop. Res.* **29**, 289–296 (2011).
55. Juneja, S. C., Schwarz, E. M., O'Keefe, R. J. & Awad, H. A. Cellular and molecular factors in flexor tendon repair and adhesions: a histological and gene expression analysis. *Connect. Tissue Res.* **54**, 218–226 (2013).
56. Chhabra, A. et al. GDF-5 deficiency in mice delays Achilles tendon healing. *J. Orthop. Res.* **21**, 826–835 (2003).
57. Katzel, E. B. et al. Impact of Smad3 loss of function on scarring and adhesion formation during tendon healing. *J. Orthop. Res.* **29**, 684–693 (2011).
58. Docheva, D., Hunziker, E. B., Fassler, R. & Brandau, O. Tenomodulin is necessary for tenocyte proliferation and tendon maturation. *Mol. Cell Biol.* **25**, 699–705 (2005).
59. Shukunami, C., Takimoto, A., Oro, M. & Hiraki, Y. Scleraxis positively regulates the expression of tenomodulin, a differentiation marker of tenocytes. *Dev. Biol.* **298**, 234–247 (2006).
60. Carmeliet, P. et al. Impaired myocardial angiogenesis and ischemic cardiomyopathy in mice lacking the vascular endothelial growth factor isoforms VEGF164 and VEGF188. *Nat. Med.* **5**, 495–502 (1999).
61. Hellstrom, M., Kalen, M., Lindahl, P., Abramsson, A. & Betsholtz, C. Role of PDGF-B and PDGFR- β in recruitment of vascular smooth muscle cells and pericytes during embryonic

- blood vessel formation in the mouse. *Development* **126**, 3047–3055 (1999).
62. Hee, C. K., Dines, J. S., Solchaga, L. A., Shah, V. R. & Hollinger, J. O. Regenerative tendon and ligament healing: opportunities with recombinant human platelet-derived growth factor BB-homodimer. *Tissue Eng. Part B Rev.* **18**, 225–234 (2012).
63. Kaux, J. F. Vascular endothelial growth factor-111 (VEGF-111) and tendon healing: preliminary results in a rat model of tendon injury. *Muscles Ligaments Tendons J.* **4**, 24–28 (2014).
64. Shah, P., Keppler, L. & Rutkowski, J. A review of platelet derived growth factor playing pivotal role in bone regeneration. *J. Oral Implantol.* **40**, 330–340 (2014).
65. Lupu, F., Terwilliger, J. D., Lee, K., Segre, G. V. & Efstratiadis, A. Roles of growth hormone and insulin-like growth factor 1 in mouse postnatal growth. *Dev. Biol.* **229**, 141–162 (2001).
66. Hansen, M. et al. Local administration of insulin-like growth factor-I (IGF-I) stimulates tendon collagen synthesis in humans. *Scand. J. Med. Sci. Sports* **23**, 614–619 (2013).
67. Nielsen, R. H. et al. Chronic alterations in growth hormone/insulin-like growth factor-I signaling lead to changes in mouse tendon structure. *Matrix Biol.* **34**, 96–104 (2014).
68. Herchenhan, A., Bayer, M. L., Eliasson, P., Magnusson, S. P. & Kjaer, M. Insulin-like growth factor I enhances collagen synthesis in engineered human tendon tissue. *Growth Horm. IGF Res.* **25**, 13–19 (2015).
69. Hagerty, P. et al. The effect of growth factors on both collagen synthesis and tensile strength of engineered human ligaments. *Biomaterials* **33**, 6355–6361 (2012).
70. Dahlgren, L. A., van der Meulen, M. C., Bertram, J. E., Starak, G. S. & Nixon, A. J. Insulin-like growth factor-I improves cellular and molecular aspects of healing in a collagenase-induced model of flexor tendinitis. *J. Orthop. Res.* **20**, 910–919 (2002).
71. Cucchiari, M. & Madry, H. Overexpression of human IGF-I via direct rAAV-mediated gene transfer improves the early repair of articular cartilage defects *in vivo*. *Gene Ther.* **21**, 811–819 (2014).
72. Schweitzer, R., Zelzer, E. & Volk, T. Connecting muscles to tendons: tendons and musculoskeletal development in flies and vertebrates. *Development* **137**, 2807–2817 (2010).
73. Havis, E. et al. Transcriptomic analysis of mouse limb tendon cells during development. *Development* **141**, 3683–3696 (2014).
74. Brown, J. P., Finley, V. G. & Kuo, C. K. Embryonic mechanical and soluble cues regulate tendon progenitor cell gene expression as a function of developmental stage and anatomical origin. *J. Biomech.* **47**, 214–222 (2014).
75. Goncalves, A. I. et al. Understanding the role of growth factors in modulating stem cell tenogenesis. *PLoS ONE* **8**, e83734 (2013).
76. Barsby, T. & Guest, D. Transforming growth factor β 3 promotes tendon differentiation of equine embryo-derived stem cells. *Tissue Eng. Part A* **19**, 2156–2165 (2013).
77. Kapacee, Z. et al. Synthesis of embryonic tendon-like tissue by human marrow stromal/mesenchymal stem cells requires a three-dimensional environment and transforming growth factor β 3. *Matrix Biol.* **29**, 668–677 (2010).
78. Farhat, Y. M. et al. Gene expression analysis of the pleiotropic effects of TGF- β 1 in an *in vitro* model of flexor tendon healing. *PLoS ONE* **7**, e51411 (2012).
79. Barsby, T., Bavin, E. P. & Guest, D. J. Three-dimensional culture and transforming growth factor β 3 synergistically promote tenogenic differentiation of equine embryo-derived stem cells. *Tissue Eng. Part A* **20**, 2604–2613 (2014).
80. Bayer, M. L. et al. Release of tensile strain on engineered human tendon tissue disturbs cell adhesions, changes matrix architecture, and induces an inflammatory phenotype. *PLoS ONE* **9**, e86078 (2014).
81. Majewski, M. et al. Improvement of tendon repair using muscle grafts transduced with TGF- β 1 cDNA. *Eur. Cell. Mater.* **23**, 94–101 (2012).
82. Mendias, C. L., Bakhurin, K. I. & Faulkner, J. A. Tendons of myostatin-deficient mice are small, brittle, and hypocellular. *Proc. Natl Acad. Sci. USA* **105**, 388–393 (2008).
83. Massague, J. TGF β signalling in context. *Nat. Rev. Mol. Cell Biol.* **13**, 616–630 (2012).
84. Berthet, E. et al. Smad3 binds Scleraxis and Mohawk and regulates tendon matrix organization. *J. Orthop. Res.* **31**, 1475–1483 (2013).
85. Lee, J. Y. et al. BMP-12 treatment of adult mesenchymal stem cells *in vitro* augments tendon-like tissue formation and defect repair *in vivo*. *PLoS ONE* **6**, e17531 (2011).
86. Park, A. et al. Adipose-derived mesenchymal stem cells treated with growth differentiation factor-5 express tendon-specific markers. *Tissue Eng. Part A* **16**, 2941–2951 (2010).
87. Hoffmann, A. et al. Neotendon formation induced by manipulation of the Smad8 signalling pathway in mesenchymal stem cells. *J. Clin. Invest.* **116**, 940–952 (2006).
88. James, R., Kumbhar, S. G., Laurencin, C. T., Balian, G. & Chhabra, A. B. Tendon tissue engineering: adipose-derived stem cell and GDF-5 mediated regeneration using electrospun matrix systems. *Biomed. Mater.* **6**, 025011 (2011).
89. Majewski, M. et al. Ex vivo adenoviral transfer of bone morphogenetic protein 12 (BMP-12) cDNA improves Achilles tendon healing in a rat model. *Gene Ther.* **15**, 1139–1146 (2008).
90. Pelled, G. et al. Smad8/BMP2-engineered mesenchymal stem cells induce accelerated recovery of the biomechanical properties of the Achilles tendon. *J. Orthop. Res.* **30**, 1932–1939 (2012).
91. Mazerbourg, S. et al. Identification of receptors and signaling pathways for orphan bone morphogenetic protein/growth differentiation factor ligands based on genomic analyses. *J. Biol. Chem.* **280**, 32122–32132 (2005).
92. Storm, E. E. et al. Limb alterations in brachypodism mice due to mutations in a new member of the TGF β -superfamily. *Nature* **368**, 639–643 (1994).
93. Settle, S. H. Jr. et al. Multiple joint and skeletal patterning defects caused by single and double mutations in the mouse *Gdf6* and *Gdf5* genes. *Dev. Biol.* **254**, 116–130 (2003).
94. Lee, K. J., Mendelsohn, M. & Jessell, T. M. Neuronal patterning by BMPs: a requirement for GDF7 in the generation of a discrete class of commissural interneurons in the mouse spinal cord. *Genes Dev.* **12**, 3394–3407 (1998).
95. Settle, S. et al. The BMP family member *Gdf7* is required for seminal vesicle growth, branching morphogenesis, and cytodifferentiation. *Dev. Biol.* **234**, 138–150 (2001).
96. Mikic, B., Schalet, B. J., Clark, R. T., Gaschen, V. & Hunziker, E. B. GDF-5 deficiency in mice alters the ultrastructure, mechanical properties and composition of the Achilles tendon. *J. Orthop. Res.* **19**, 365–371 (2001).
97. Mikic, B., Bierwert, L. & Tsou, D. Achilles tendon characterization in GDF-7 deficient mice. *J. Orthop. Res.* **24**, 831–841 (2006).
98. Clark, R. T. et al. GDF-5 deficiency in mice leads to disruption of tail tendon form and function. *Connect. Tissue Res.* **42**, 175–186 (2001).
99. Blitz, E. et al. Bone ridge patterning during musculoskeletal assembly is mediated through SCX regulation of *Bmp4* at the tendon-skeleton junction. *Dev. Cell* **17**, 861–873 (2009).
100. Blitz, E., Sharir, A., Akiyama, H. & Zelzer, E. Tendon-bone attachment unit is formed modularly by a distinct pool of Scx- and Sox9-positive progenitors. *Development* **140**, 2680–2690 (2013).
101. Hogan, M. et al. Growth differentiation factor-5 regulation of extracellular matrix gene expression in murine tendon fibroblasts. *J. Tissue Eng. Regen. Med.* **5**, 191–200 (2011).
102. Edom-Vovard, F., Schuler, B., Bonnin, M. A., Teillet, M. A. & Duprez, D. *Fgf4* positively regulates scleraxis and tenascin expression in chick limb tendons. *Dev. Biol.* **247**, 351–366 (2002).
103. Brent, A. E., Schweitzer, R. & Tabin, C. J. A somitic compartment of tendon progenitors. *Cell* **113**, 235–248 (2003).
104. Brent, A. E. & Tabin, C. J. FGF acts directly on the somitic tendon progenitors through the Ets transcription factors *Pea3* and *Erm* to regulate scleraxis expression. *Development* **131**, 3885–3896 (2004).
105. Eloy-Trinquet, S., Wang, H., Edom-Vovard, F. & Duprez, D. Fgf signaling components are associated with muscles and tendons during limb development. *Dev. Dyn.* **238**, 1195–1206 (2009).
106. Paxton, J. Z., Hagerty, P., Andrick, J. J. & Baar, K. Optimizing an intermittent stretch paradigm using ERK1/2 phosphorylation results in increased collagen synthesis in engineered ligaments. *Tissue Eng. Part A* **18**, 277–284 (2012).
107. Thomopoulos, S. et al. The effects of exogenous basic fibroblast growth factor on intrasynovial flexor tendon healing in a canine model. *J. Bone Joint Surg. Am.* **92**, 2285–2293 (2010).
108. Tang, J. B. et al. Adeno-associated virus-2-mediated bFGF gene transfer to digital flexor tendons significantly increases healing strength: an *in vivo* study. *J. Bone Joint Surg. Am.* **90**, 1078–1089 (2008).
109. Tang, J. B., Chen, C. H., Zhou, Y. L., McKeever, C. & Liu, P. Y. Regulatory effects of introduction of an exogenous FGF2 gene on other growth factor genes in a healing tendon. *Wound Repair Regen.* **22**, 111–118 (2014).
110. Baksh, N., Hannon, C. P., Murawski, C. D., Smyth, N. A. & Kennedy, J. G. Platelet-rich plasma in tendon models: a systematic review of basic science literature. *Arthroscopy* **29**, 596–607 (2013).
111. Kaux, J. F. Comparative study of five techniques of preparation of platelet-rich plasma [French]. *Pathol. Biol. (Paris)* **59**, 157–160 (2011).
112. Alberton, P. et al. Conversion of human bone marrow-derived mesenchymal stem cells into tendon progenitor cells by ectopic expression of scleraxis. *Stem Cells Dev.* **21**, 846–858 (2012).
113. Chen, X. et al. Scleraxis-overexpressed human embryonic stem cell-derived mesenchymal stem cells for tendon tissue engineering with knitted silk-collagen scaffold. *Tissue Eng. Part A* **20**, 1583–1592 (2014).
114. Tan, C., Lui, P. P., Lee, Y. W. & Wong, Y. M. Scx-transduced tendon-derived stem cells (TDSCs) promoted better tendon repair compared to mock-transduced cells in a rat patellar tendon window injury model. *PLoS ONE* **9**, e97453 (2014).

115. Liu, H. *et al.* Mohawk promotes the tenogenesis of mesenchymal stem cells through activation of the TGF β signaling pathway. *Stem Cells* **33**, 443–455 (2015).
116. Otabe, K. *et al.* Transcription factor Mohawk controls tenogenic differentiation of bone marrow mesenchymal stem cells *in vitro* and *in vivo*. *J. Orthop. Res.* **33**, 1–8 (2015).
117. Kardon, G. Muscle and tendon morphogenesis in the avian hind limb. *Development* **125**, 4019–4032 (1998).
118. Schweitzer, R. *et al.* Analysis of the tendon cell fate using Scleraxis, a specific marker for tendons and ligaments. *Development* **128**, 3855–3866 (2001).
119. Bonnin, M. A. *et al.* *Shh* is not involved in limb tendon development, but is expressed in limb connective tissue under *Shh* regulation. *Mech. Dev.* **122**, 573–585 (2005).
120. Maeda, T. *et al.* Conversion of mechanical force into TGF β -mediated biochemical signals. *Curr. Biol.* **21**, 933–941 (2011).
121. Heinemeier, K. M. *et al.* Effect of unloading followed by reloading on expression of collagen and related growth factors in rat tendon and muscle. *J. Appl. Physiol.* (1985) **106**, 178–186 (2009).
122. de Boer, M. D. *et al.* The temporal responses of protein synthesis, gene expression and cell signalling in human quadriceps muscle and patellar tendon to disuse. *J. Physiol.* **585**, 241–251 (2007).
123. Mendias, C. L., Gumucio, J. P., Bakhurin, K. I., Lynch, E. B. & Brooks, S. V. Physiological loading of tendons induces scleraxis expression in epitenon fibroblasts. *J. Orthop. Res.* **30**, 606–612 (2012).
124. Chen, J. L. *et al.* Efficacy of hESC-MSCs in knitted silk-collagen scaffold for tendon tissue engineering and their roles. *Biomaterials* **31**, 9438–9451 (2010).
125. Scott, A. *et al.* Mechanical force modulates scleraxis expression in bioartificial tendons. *J. Musculoskelet. Neuronal Interact.* **11**, 124–132 (2011).
126. Morita, Y., Watanabe, S., Ju, Y. & Xu, B. Determination of optimal cyclic uniaxial stretches for stem cell-to-tenocyte differentiation under a wide range of mechanical stretch conditions by evaluating gene expression and protein synthesis levels. *Acta Bioeng. Biomech.* **15**, 71–79 (2013).
127. Killian, M. L., Cavinatto, L., Galatz, L. M. & Thomopoulos, S. The role of mechanobiology in tendon healing. *J. Shoulder Elbow Surg.* **21**, 228–237 (2012).
128. Eliasson, P., Andersson, T. & Aspenberg, P. Achilles tendon healing in rats is improved by intermittent mechanical loading during the inflammatory phase. *J. Orthop. Res.* **30**, 274–279 (2012).
129. Eliasson, P., Andersson, T. & Aspenberg, P. Rat Achilles tendon healing: mechanical loading and gene expression. *J. Appl. Physiol.* (1985) **107**, 399–407 (2009).
130. Andersson, T., Eliasson, P., Hammerman, M., Sandberg, O. & Aspenberg, P. Low-level mechanical stimulation is sufficient to improve tendon healing in rats. *J. Appl. Physiol.* (1985) **113**, 1398–1402 (2012).
131. Gimbel, J. A., Van Kleunen, J. P., Williams, G. R., Thomopoulos, S. & Soslowsky, L. J. Long durations of immobilization in the rat result in enhanced mechanical properties of the healing supraspinatus tendon insertion site. *J. Biomech. Eng.* **129**, 400–404 (2007).
132. Pagel, J. I. & Deindl, E. Early growth response 1—a transcription factor in the crossfire of signal transduction cascades. *Indian J. Biochem. Biophys.* **48**, 226–235 (2011).
133. Eliasson, P., Andersson, T., Hammerman, M. & Aspenberg, P. Primary gene response to mechanical loading in healing rat Achilles tendons. *J. Appl. Physiol.* (1985) **114**, 1519–1526 (2013).
134. Hammerman, M., Aspenberg, P. & Eliasson, P. Microtrauma stimulates rat Achilles tendon healing via an early gene expression pattern similar to mechanical loading. *J. Appl. Physiol.* (1985) **116**, 54–60 (2014).
135. Matsakas, A., Otto, A., Elashry, M. I., Brown, S. C. & Patel, K. Altered primary and secondary myogenesis in the myostatin-null mouse. *Rejuvenation Res.* **13**, 717–727 (2010).
136. Hankemeier, S. *et al.* Modulation of proliferation and differentiation of human bone marrow stromal cells by fibroblast growth factor 2: potential implications for tissue engineering of tendons and ligaments. *Tissue Eng.* **11**, 41–49 (2005).
137. Wang, X. T., Liu, P. Y., Xin, K. Q. & Tang, J. B. Tendon healing *in vitro*: bFGF gene transfer to tenocytes by adeno-associated viral vectors promotes expression of collagen genes. *J. Hand Surg. Am.* **30**, 1255–1261 (2005).
138. Thomopoulos, S. *et al.* bFGF and PDGF-BB for tendon repair: controlled release and biologic activity by tendon fibroblasts *in vitro*. *Ann. Biomed. Eng.* **38**, 225–234 (2010).
139. Sahoo, S., Toh, S. L. & Goh, J. C. A bFGF-releasing silk/PLGA-based biohybrid scaffold for ligament/tendon tissue engineering using mesenchymal progenitor cells. *Biomaterials* **31**, 2990–2998 (2010).
140. Raghavan, S. S. *et al.* Optimization of human tendon tissue engineering: synergistic effects of growth factors for use in tendon scaffold repopulation. *Plast. Reconstr. Surg.* **129**, 479–489 (2012).
141. Calliari, S. R. & Harley, B. A. Composite growth factor supplementation strategies to enhance tenocyte bioactivity in aligned collagen-GAG scaffolds. *Tissue Eng. Part A* **19**, 1100–1112 (2013).

Acknowledgements

The authors thank S. Gournet for assistance with illustrations. The authors' work is supported by funding from the National Institute of Health and Medical Research (INSERM). D.D. also receives support from the Fondation pour la Recherche Médicale (FRM), Agence nationale de la recherche (ANR), Association Française contre les Myopathies, Centre national de la recherche scientifique (CNRS) and Pierre & Marie Curie University (UPMC), Fondation Arthritis Courtin and Société Française de Rhumatologie.

Author contributions

All authors contributed to researching data for the article, providing a substantial contribution to discussions of the content, writing the article, and to the review and/or editing of the manuscript before submission.